Semantic Modelling of a Histopathology Image Exploration and Analysis Tool

Defense for the Ph.D. in **Medical Informatics**

By **Lamine TRAORE**

December 8, 2017

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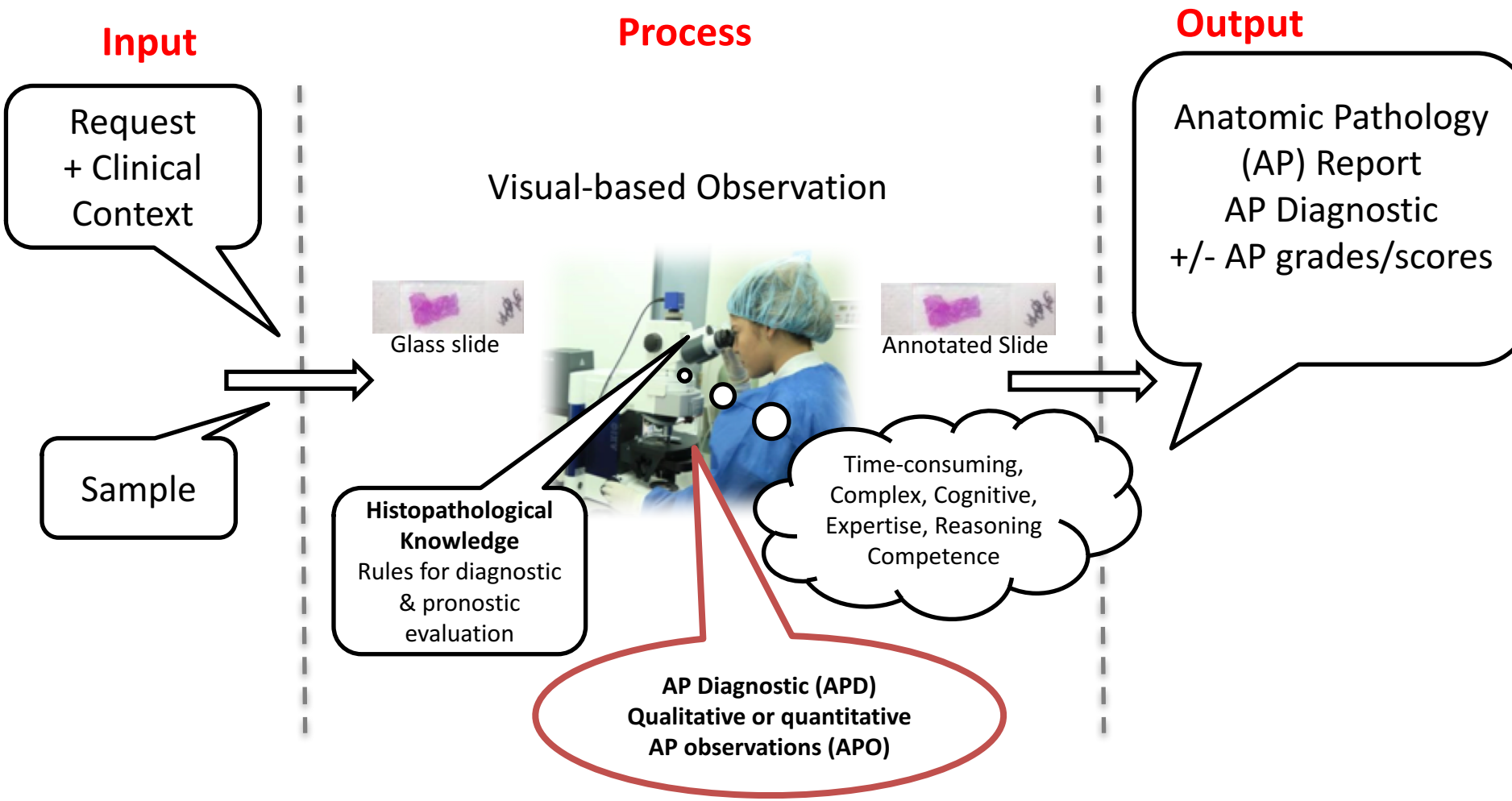
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1. Introduction

Context



Current scenario for the exploration of a Glass slide in a Anatomic Pathology Laboratory

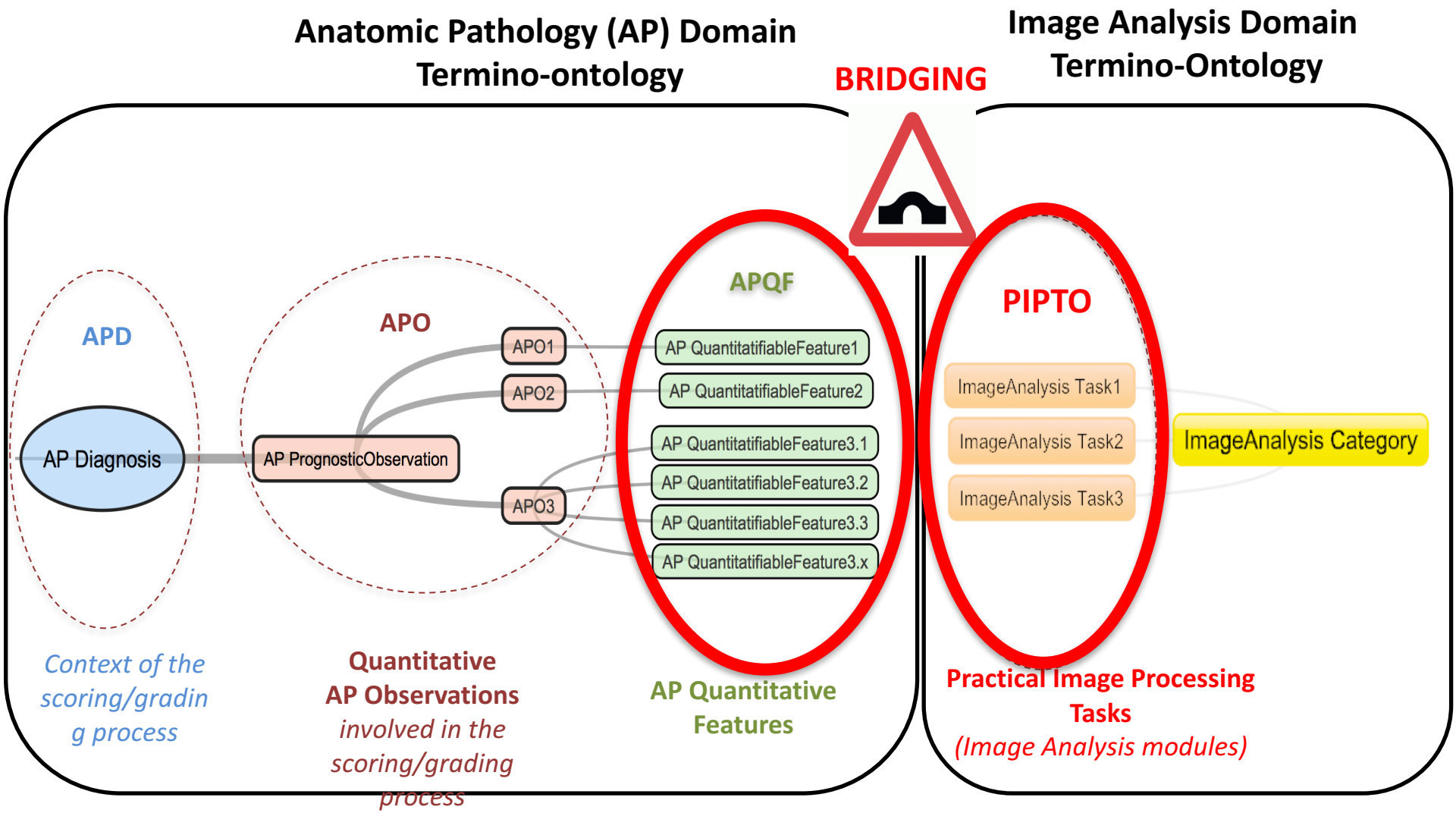
Hypothesis

A UMLS-based formal integrative representation of both quantitative **anatomic pathology (AP) observations** and **image analysis tasks** in the context of the **grading/scoring** of **malignant tumors** will contribute to

- Develop **explicit**, **unambiguous** knowledge sources for **innovative** computer assisted diagnosis systems
- Provide a knowledge base for better collaboration between **humans** and **computers** in the process of grading/scoring **malignant tumors**

In order to enhance the inter, intra **AP diagnostic reproducibility**

Objective


Anatomic Pathology (AP) Domain Termino-ontology
BRIDGING
Image Analysis Domain Termino-Ontology
APD
APO
APQF
PIPTO
AP Diagnosis
AP PrognosticObservation
APO1
APO2
APO3
AP QuantitatifiableFeature1
AP QuantitatifiableFeature2
AP QuantitatifiableFeature3.1
AP QuantitatifiableFeature3.2
AP QuantitatifiableFeature3.3
AP QuantitatifiableFeature3.x
ImageAnalysis Task1
ImageAnalysis Task2
ImageAnalysis Task3
ImageAnalysis Category
Context of the scoring/grading process
Quantitative AP Observations involved in the scoring/grading process
AP Quantitative Features
Practical Image Processing Tasks (Image Analysis modules)

Problematic & Challenge

Reproducibility & **Sustainable management** of the semantics

To **Annotate** histopathology images with labels complying with **reference vocabularies** and **semantic standards**

To **Formalize** associated knowledge for diagnostic interpretation of histopathology images by both **humans** and **machines**



2. State of Art

Existing Standards & Initiatives

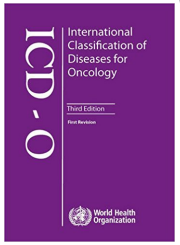
Reference Terminologies

NIH NATIONAL CANCER INSTITUTE

LOINC from Regenstrief

RSNA Informatics RadLex

SNOMED International



asipsanté L'AGENCE FRANÇAISE DE LA SANTÉ NUMÉRIQUE

ADICAP

Anatomic Pathology Lexicon

Information model & Interoperability initiatives

IHE

HL7 INTERNATIONAL

FHIR

DICOM Digital Imaging and Communications in Medicine

Literature

Zillner et al. classify patients with **lymphoma** automatically in a **spatio-anatomical context**, based on staging system, medical in annotation, **Radlex** and **FMA** **image metadata reaso** ontological model. [2012]

Racoceanu et al. d **analysis** protocol d Slide Image (WSI) **cancer** classificatio

Gurcan & Smith et al. propose **and methods** used in **pathologi** is named as « **Quantitative H** QHIO ». On-going and aims to **information-driven collaborations** in the pathological imaging field. [Feb, 2017]

→ Organ specific
→ Domain specific
→ Local knowledge source

✓ **Broad consensus with application**
✓ **Cross-disciplinary**
✓ **Reuse of existing semantic sources**

3. General Methodology

Positioning our Approach

- **Relevant application** base with the **College of American Pathologists Cancer Checklists & Protocols (CAP CC&P)**

  →broad **consensus** and ensure links to **existing standards**

- **Multidisciplinary** concern : **Pathology + Image Analysis**

  →facilitate the adoption by professionals in a larger scope

- **Reuse** of **reference ontologies** and existing **semantic sources**

  →**sustainable maintenance** of the generated knowledge, crucial in a rapidly evolving domain

MANUAL

Corpus annotation



1

Annotation by AP Expert

Identification
Quantifiable Parameters



AUTOMATIC

2

Identification by Recommender vs. original algorithms

Top **5 Reference Ontologies**

BioPortal

3

Conceptualisation by Annotator

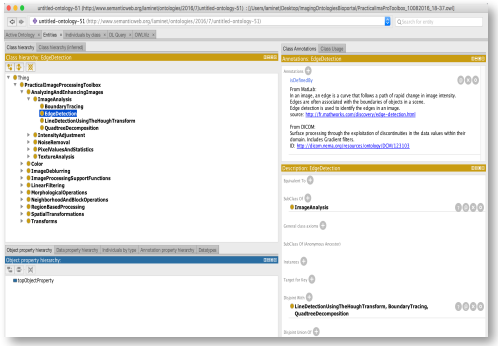
Unified Medical Language System®

Extracted Concepts

Complement Metadata by UMLS Terminology Services

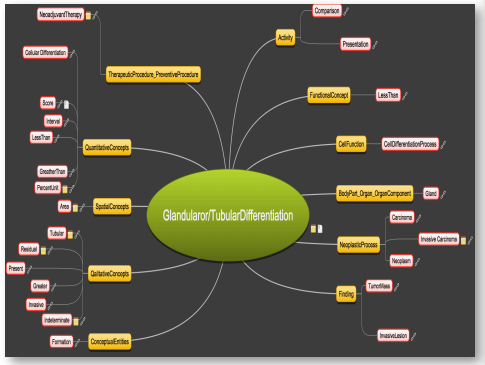
4

5



Formalisation

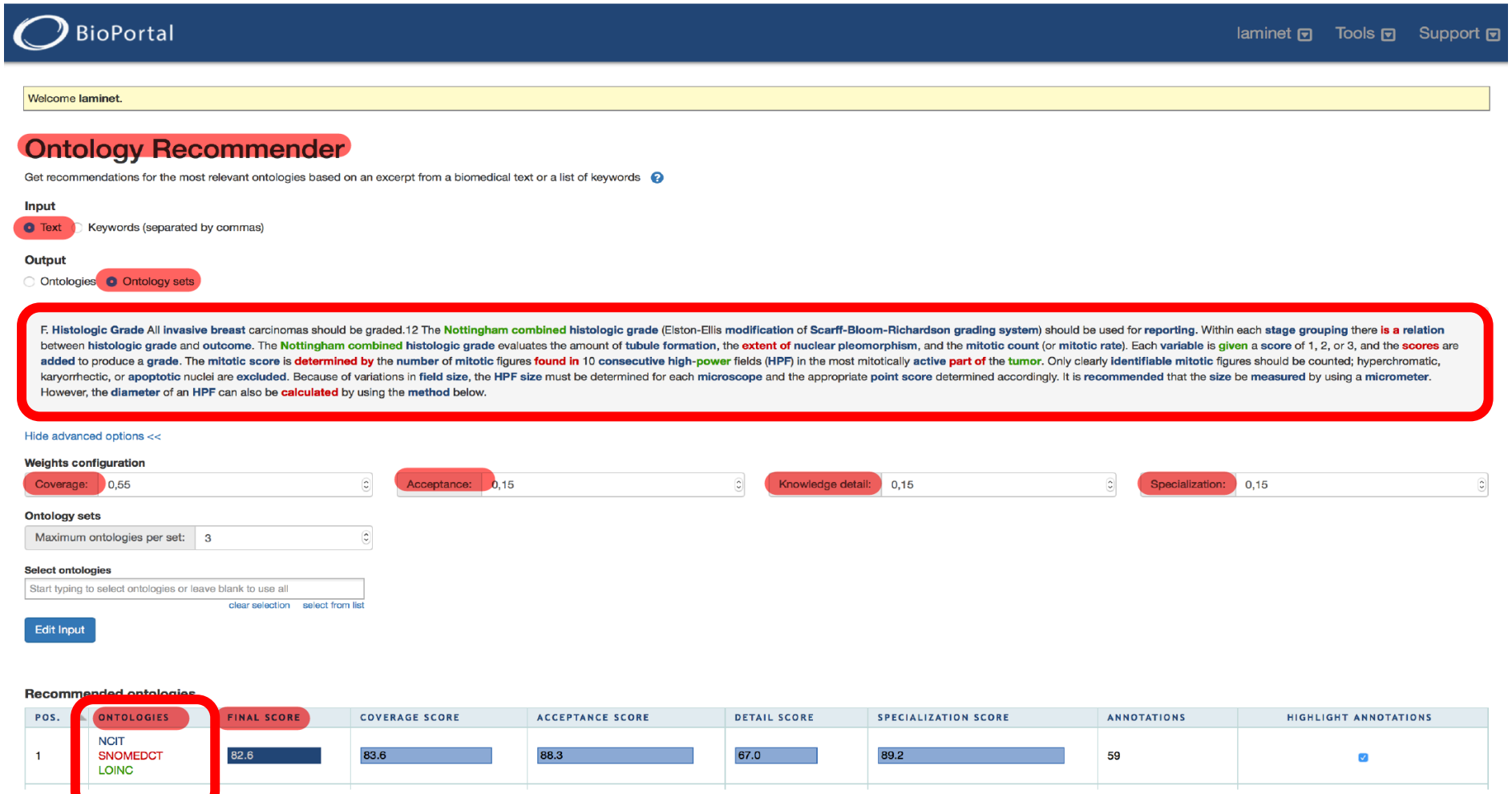
SEMI-AUTOMATIC



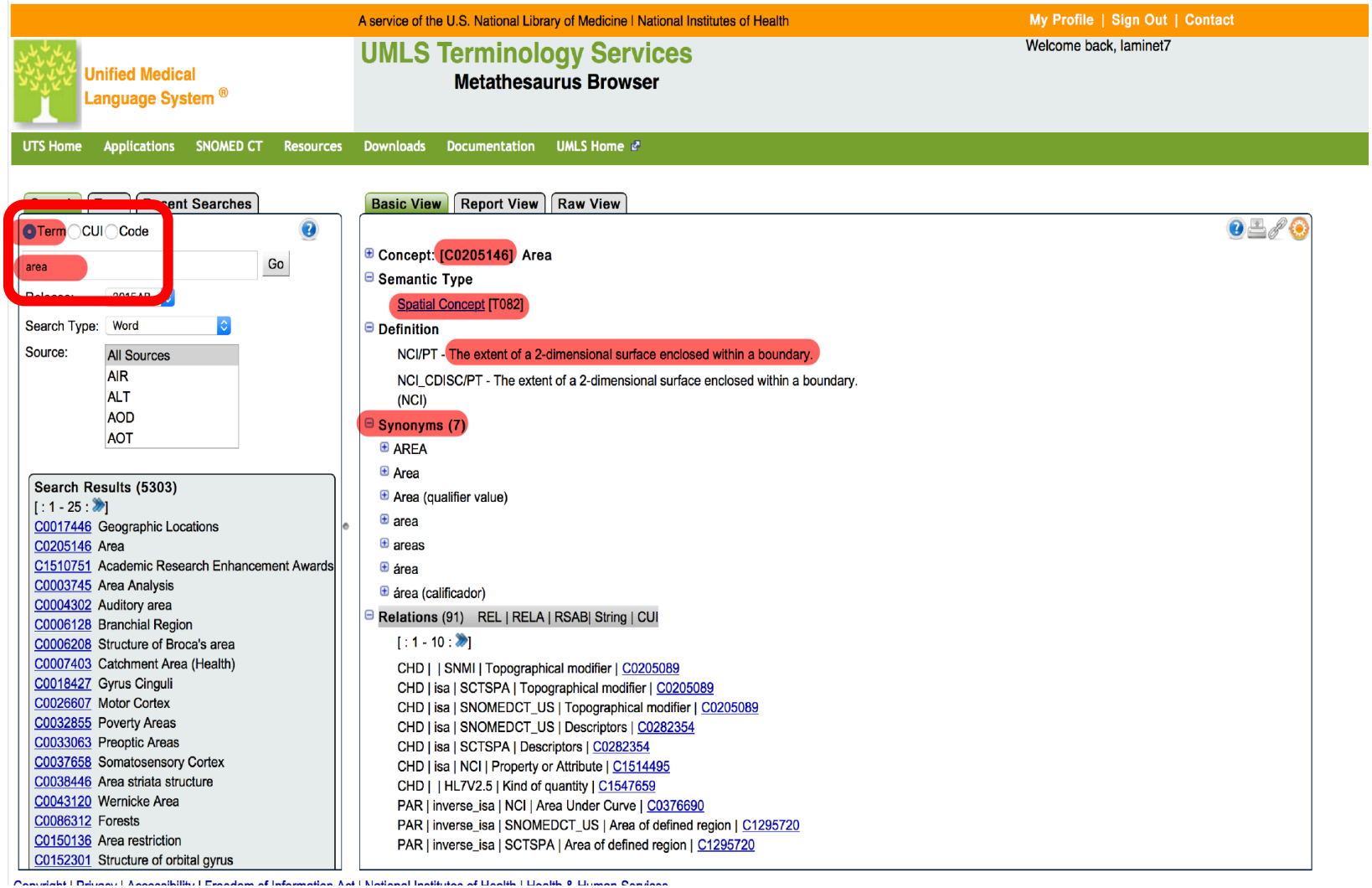
Visualisation

SEMI-AUTOMATIC / AUTOMATIC

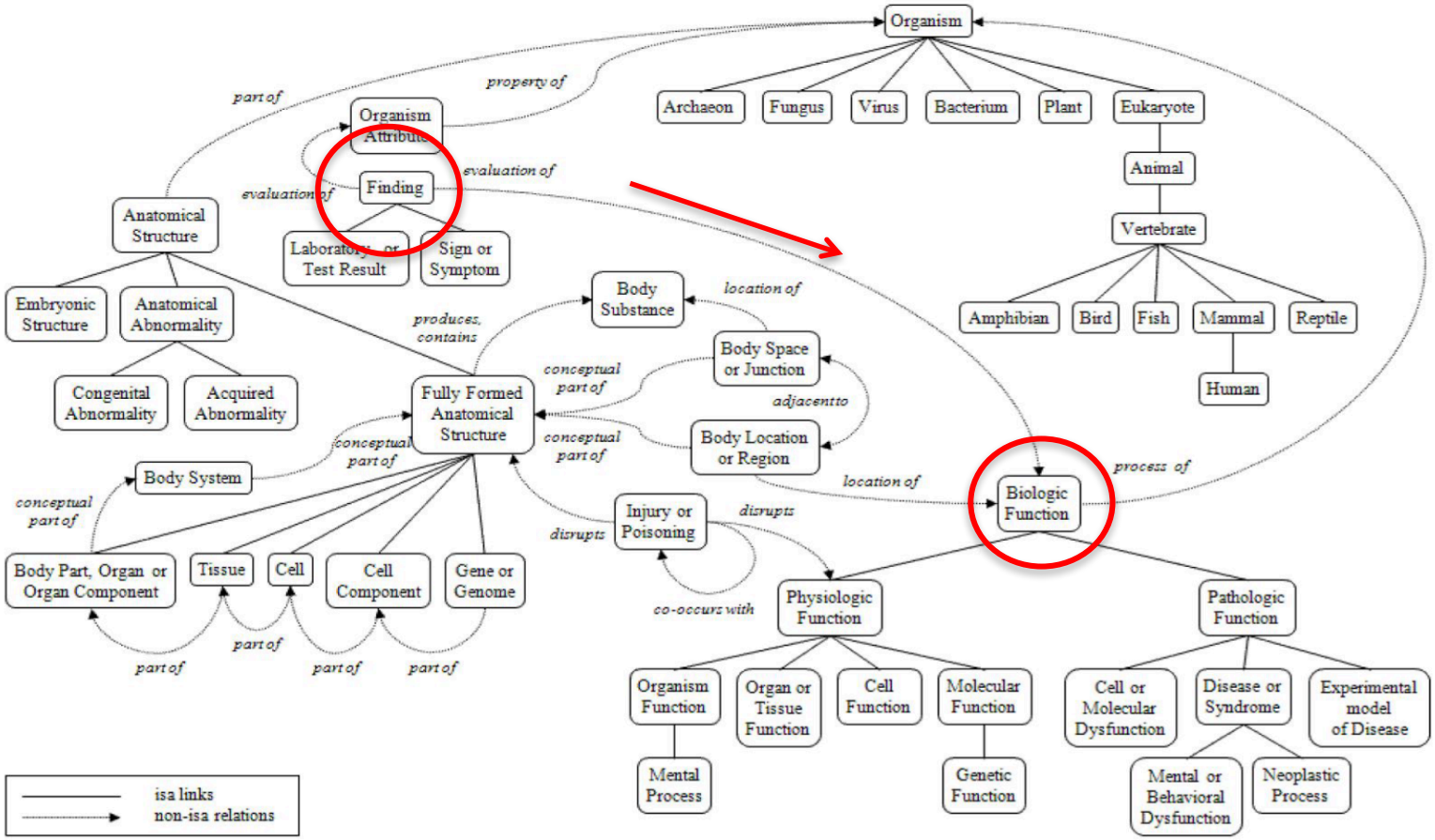
Semantic repositories & Tools (1/2)



Semantic repositories & Tools (2/2)



UMLS Metathesaurus – Semantic Types & Semantic Network



College of American Pathologists Cancer Checklists & Protocols – CAP CC&P

Protocol for the Examination of Specimens from Patients with Invasive Carcinoma of the Breast

2017 Breast v4.0.0.0

Protocolposting date: June 2017

Surgical Pathology Cancer Case Summary

INVASIVE CARCINOMA OF THE BREAST:

Procedure

Specimen Laterality

Tumor Size

Histologic Type

Histologic Grade: Nottingham Histologic Score
Glandular (Acinar)/Tubular Differentiation
Nuclear Pleomorphism
Mitotic Count
Overall Grade

Ductal Carcinoma In Situ (DCIS)

Macroscopic and Microscopic Extent of Tumor*
Skin
Nipple
Skeletal Muscle

Margins
Invasive Carcinoma*
DCIS*

Regional Lymph Nodes
Number of Lymph Nodes with Macrometastases (>2 mm): ____*
Number of Lymph Nodes with Micrometastases (>0.2 mm to 2 mm and/or >200 ce
Number of Lymph Nodes with Isolated Tumor Cells (≤0.2 mm and ≤200 cells)#: __
Number of Lymph Nodes Examined*
Number of Sentinel Nodes Examined*

Pathologic Stage Classification (pTNM, AJCC 8th Edition)
TNM Descriptors*
Primary Tumor (pT)
Regional Lymph Nodes (pN)
Distant Metastasis (pM)*

Glandularor/TubularDiffe

All invasive breast carcinomas should be graded. The Nottingham combined histologic grade (Elston-Ellis modification of Scarff-Bloom-Richardson grading system) should be used for reporting. Within each stage grouping there is a relation between histologic grade and outcome.The Nottingham combined histologic grade evaluates the amount of tubule formation, the extent of nuclear pleomorphism, and the mitotic count (or mitotic rate). Each variable is given a score of 1, 2, or 3, and the scores are added to produce a grade.

Glandular (Acinar)/Tubular Differentiation
Score 1: greater than 75 percent of tumor area forming glandular/tubular structures
Score 2: 10 to 75 percent of tumor area forming glandular/ tubular structures
Score 3: less than 10 percent of tumor area forming glandular/tubular structures
Only microinvasion present (not graded)
No residual invasive carcinoma after presurgical

Imaging corpus issued from contest

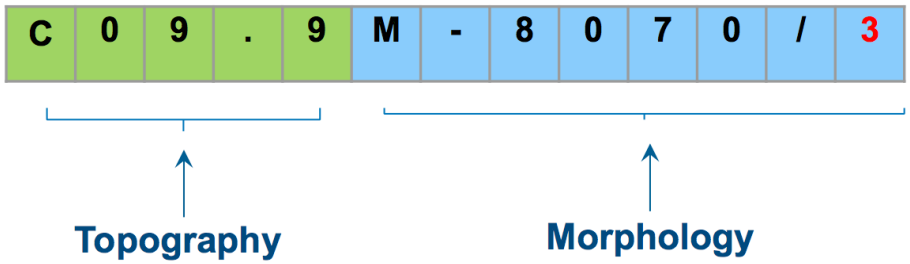
Conference -> challenge->winners->methods->articles->extracted corpus *

Corpus index	Associated conference	Selected challenges	# of methods	Word counts
C#1	ICPR 2012	MITOSIS (Mitosis detection in breast cancer histological images)	4	181
C#2	MICCAI 2013	AMIDA (Assessment of algorithms for mitosis detection in breast cancer histopathology images)	11	405
C#3	ICPR 2014	MITOS-ATYPIA (Detection of mitosis and high-grade atypia nuclei in breast cancer histology images)	4	627
C#4	MICCAI 2015	GlaS (Gland Segmentation in Colon Histology Images)	6	501
C#5	ISBI 2016	Camelyon16 (cancer metastasis detection in lymph node)	4	896
TOTAL		**5 International benchmarking Challenges**	**29 Top ranking Methods**	

***Source:** "grand-challenges - Home." [Online]. Available: https://grand-challenge.org/. [Accessed: 25-Oct-2016].

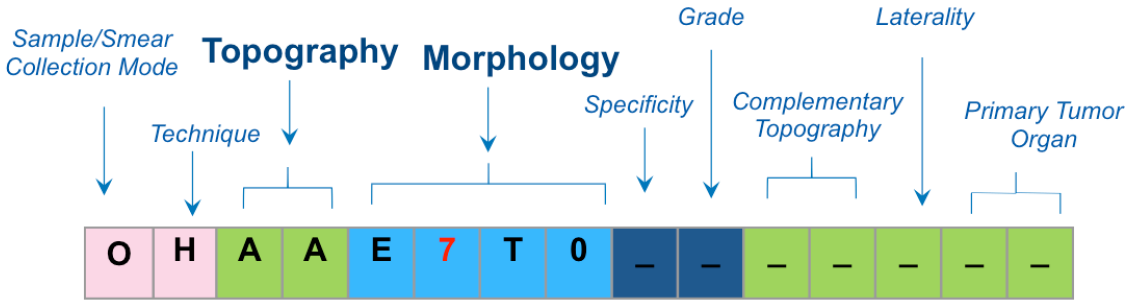
4. Results

AP Diagnosis (APD) of tumor pathology



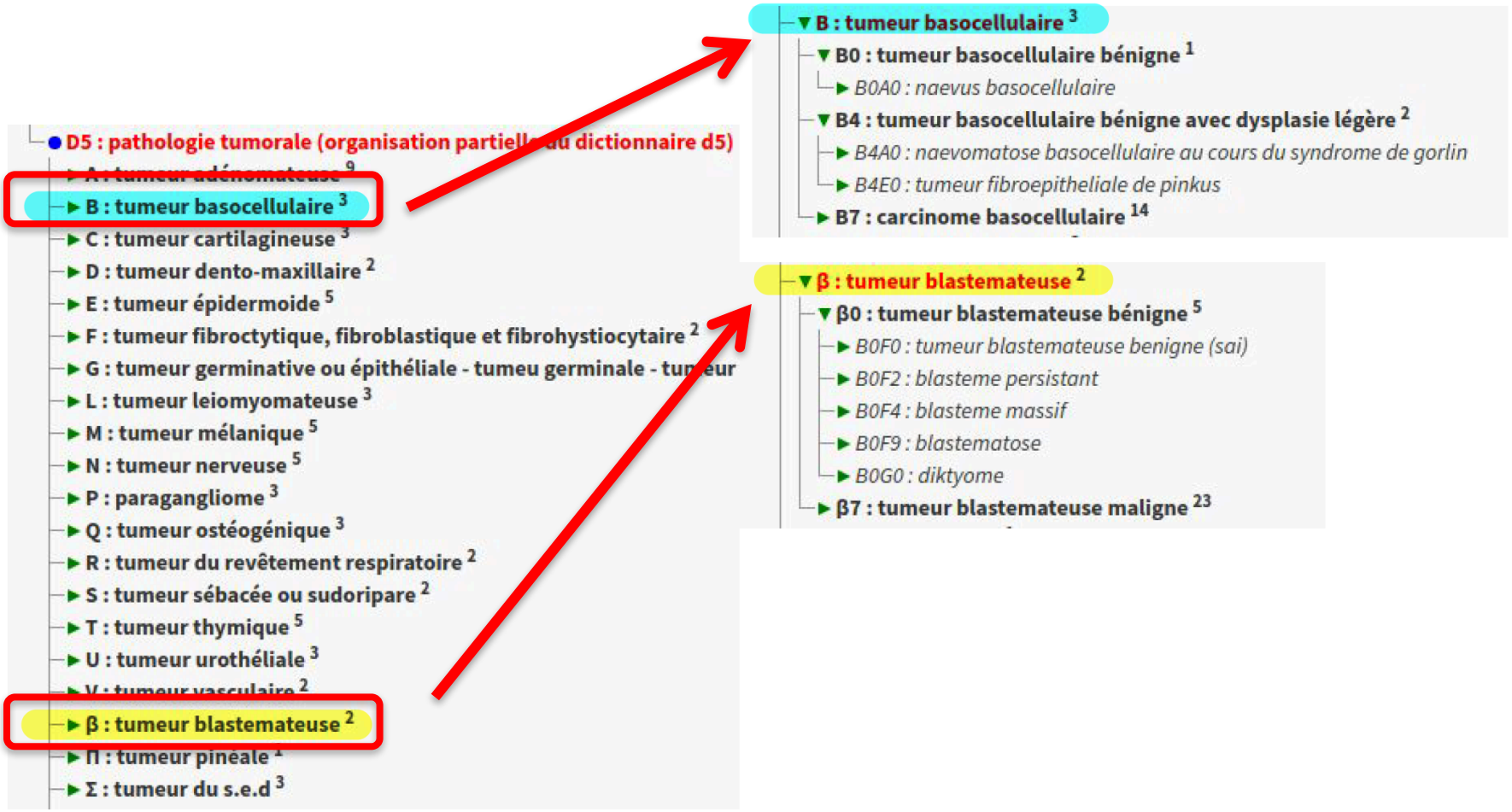


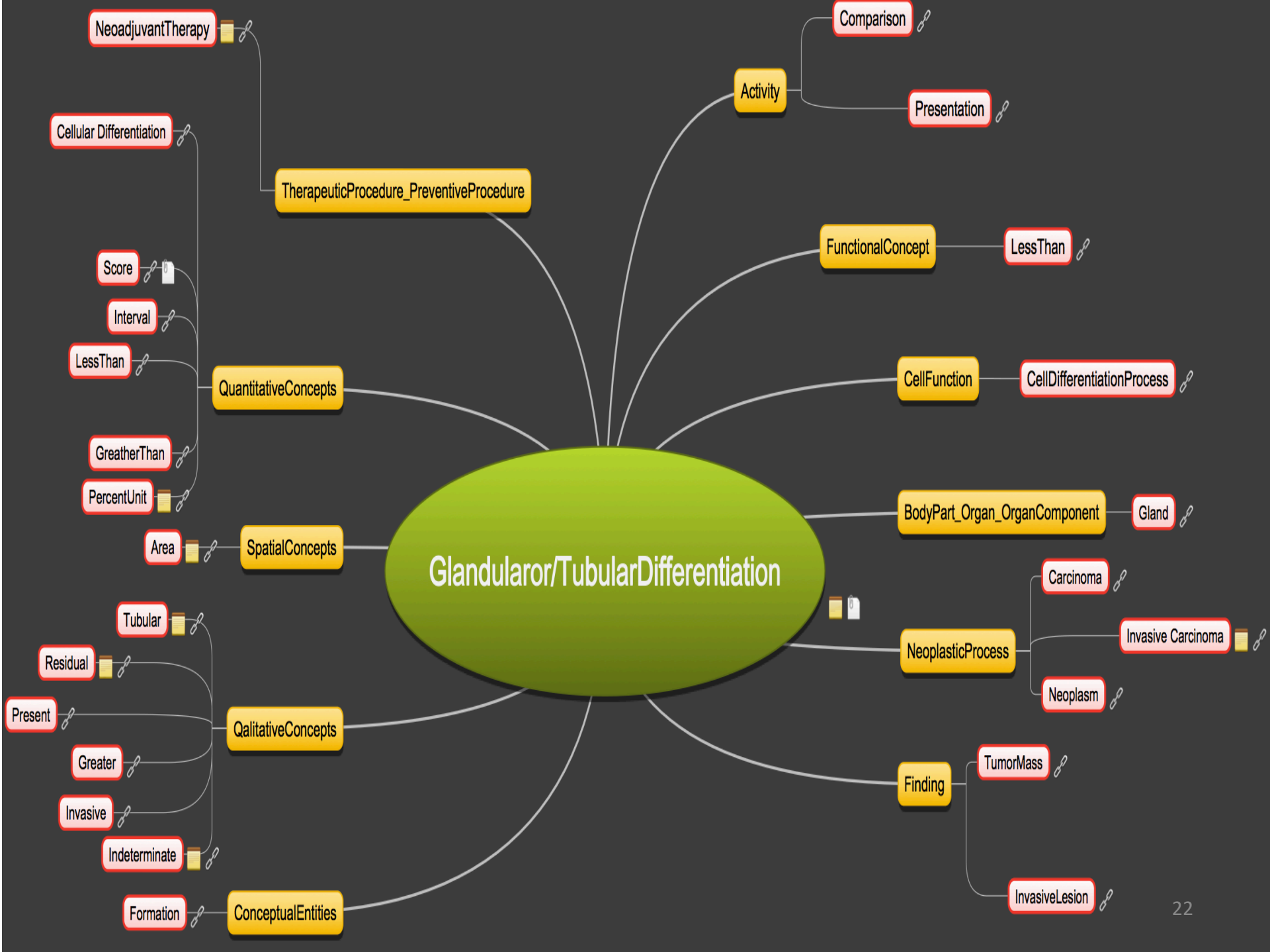
ADICAP

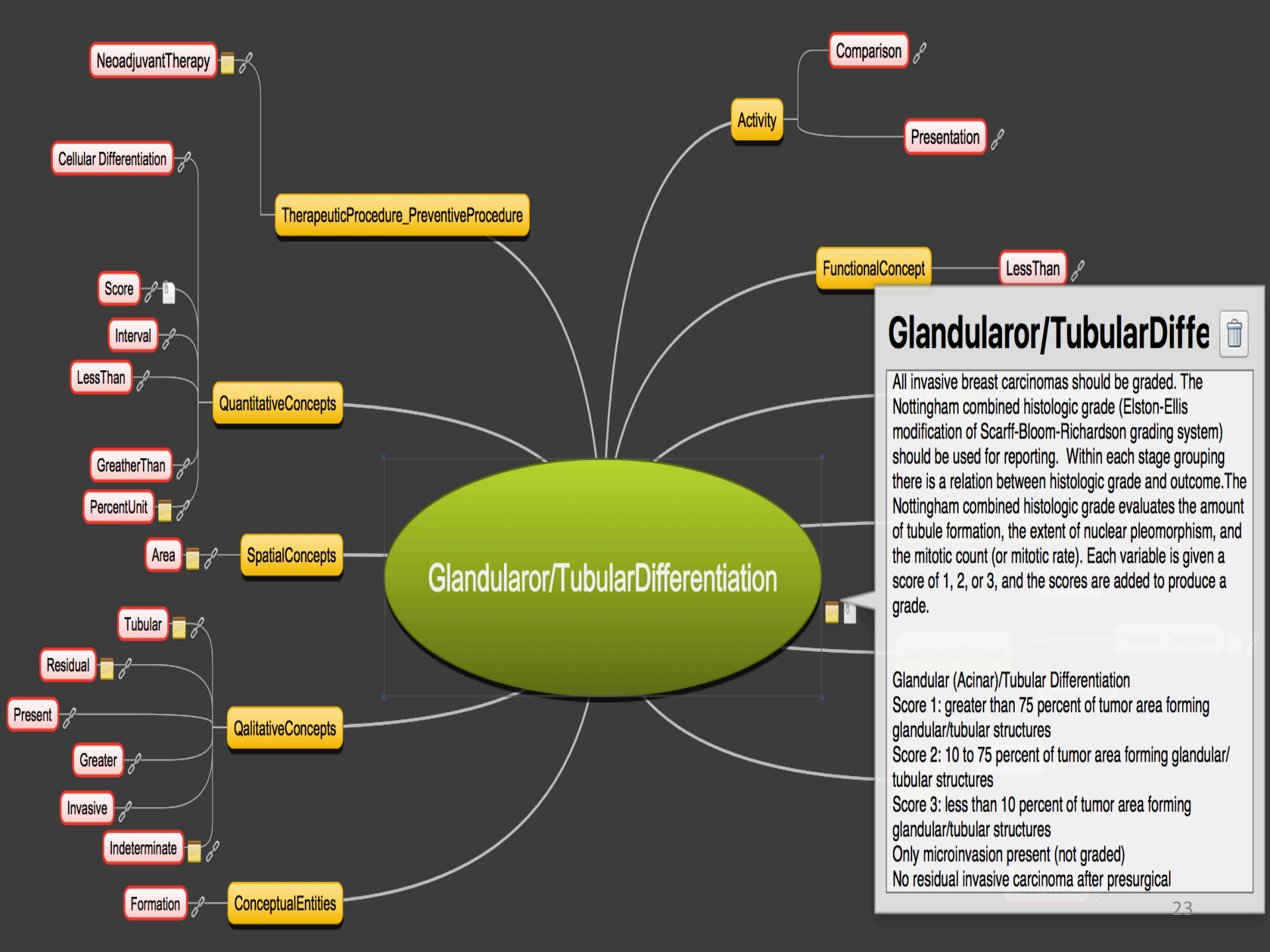


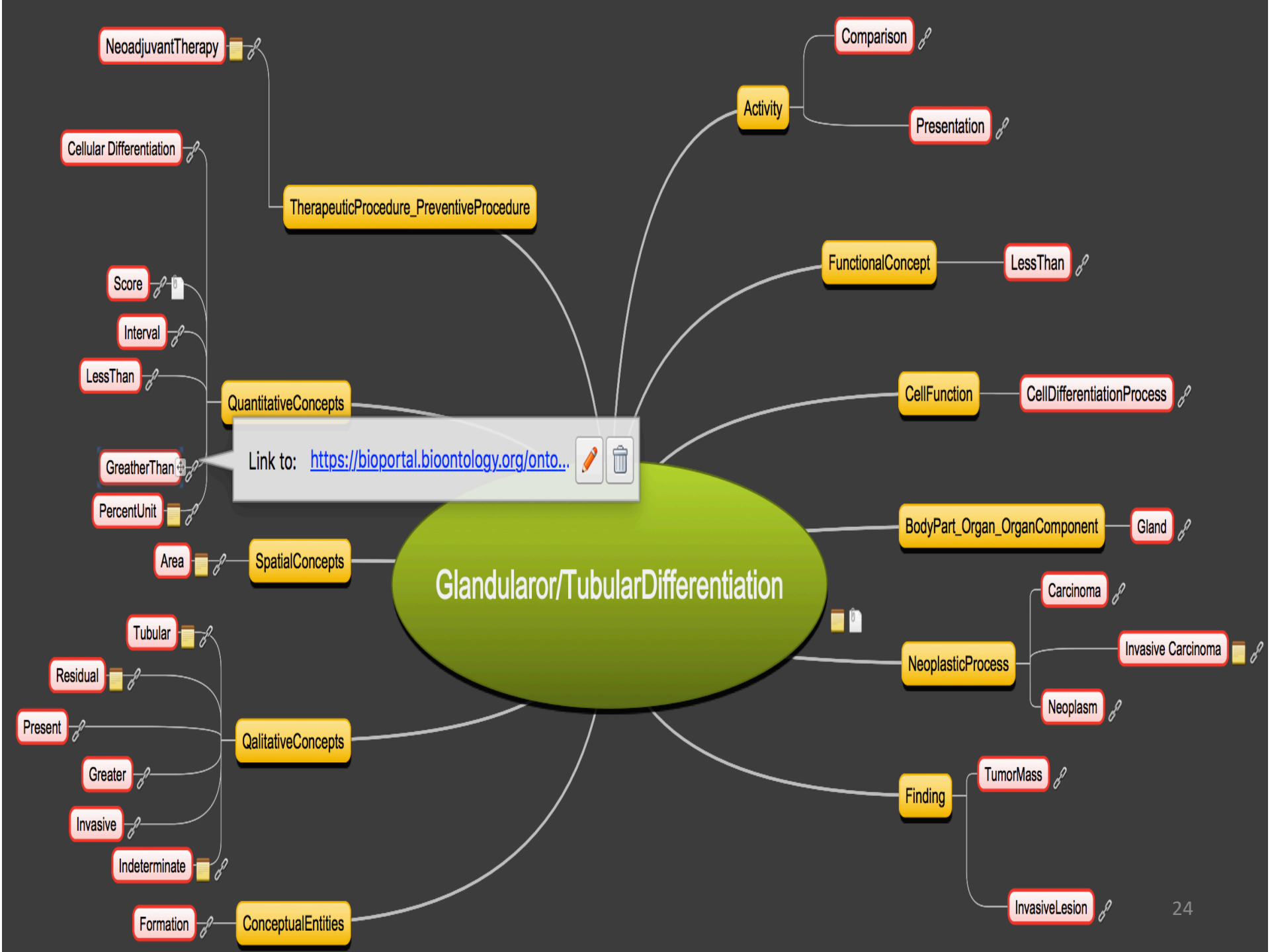
A Tumeur Adénomateuse - Adénocarcinome
B Tumeur Basocellulaire - tumeur Blastémateuse
C Tumeur Cartilagineuse
D Tumeur Dento-maxillaire - tumeur Dysgénétique
E Tumeur Epidermoide
F Tumeur Fibrocytique, Fibroblastique et Fibrohistiocytaire
G Tumeur Germinative ou épithéliale - tumeur germinale, tumeur gonadique
H,J,K Tumeur et maladie de système des tissus Hématopoïétiques
- H Tumeur et maladie de système des tissus Hématopoïétiques
- J Lymphomes (Classification de Kiel actualisée - 1988) puis OMS2000
- K Lymphomes hodgkinien et non hodgkinien (WForm ne plus utiliser)

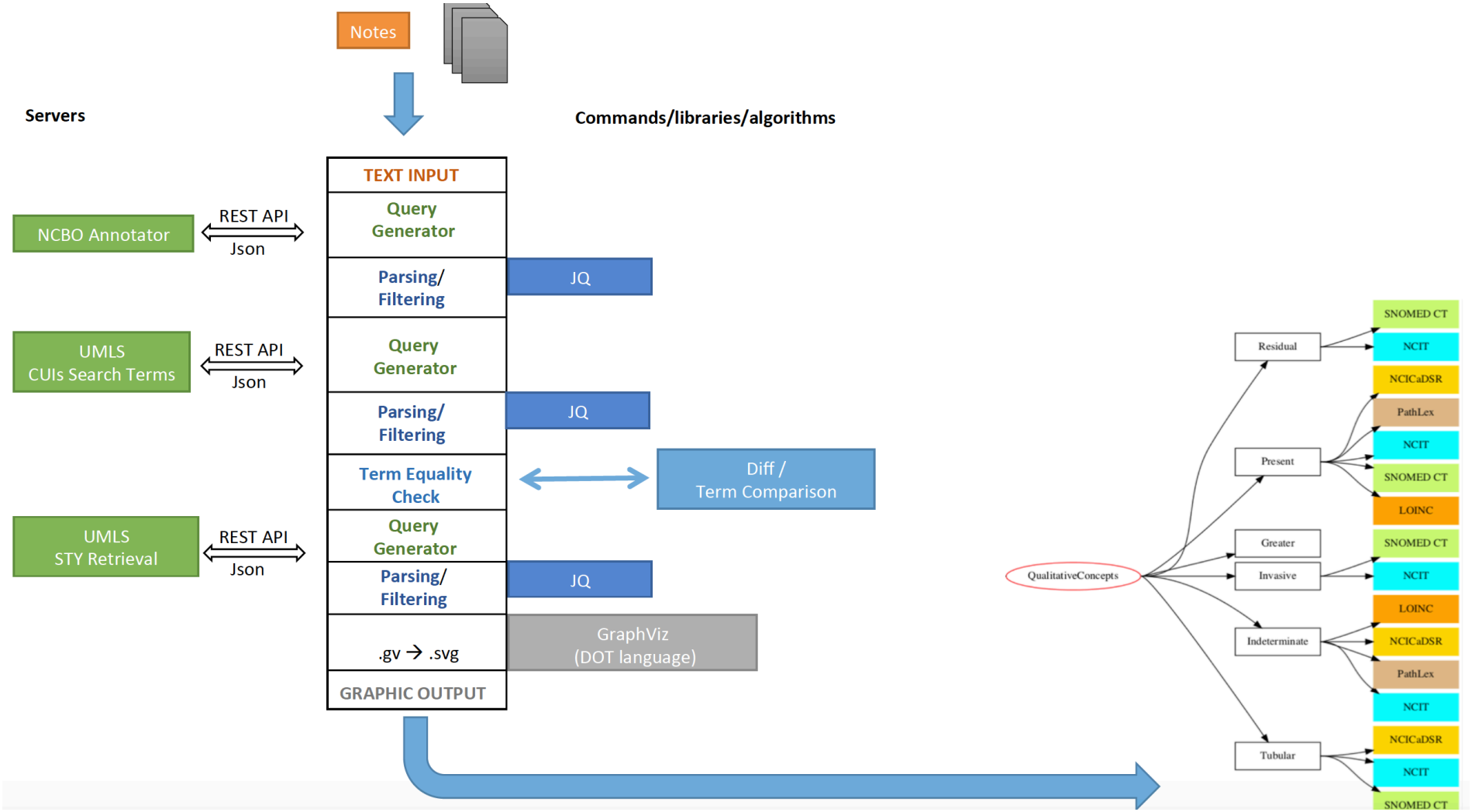
L Tumeur Léiomyomateuse - tumeur Lipomateuse
M Tumeur Mélanique - tumeur du Mésenchyme pluripotentiel, tumeur Mésoth
N Tumeur Nerveuse
P Paragangliome - tumeur pilaire- tumeur pinéale
Q Tumeur ostéogénique
R Tumeur du revêtement Respiratoire - tumeur rhabdomyomateuse
S Tumeur Sébacée, sudoripare, synoviale et du S.E.D.
T Tumeur Thymique - tumeur Trophoblastique
U Tumeur Urothéliale
V Tumeur Vasculaire
X Tumeur d'histogenèse indéterminée ou particulière


D5 : pathologie tumorale (organisation partielle du dictionnaire d5)
B : tumeur basocellulaire 3
C : tumeur cartilagineuse 3
D : tumeur dento-maxillaire 2
E : tumeur épidermoide 5
F : tumeur fibroctytique, fibroblastique et fibrohystiocytaire 2
G : tumeur germinative ou épithéliale - tumeu germinale - tu eur
L : tumeur leiomyomateuse 3
M : tumeur mélanique 5
N : tumeur nerveuse 5
P : paragangliome 3
Q : tumeur ostéogénique 3
R : tumeur du revêtement respiratoire 2
S : tumeur sébacée ou sudoripare 2
T : tumeur thymique 5
U : tumeur urothéliale 3
β : tumeur blastemateuse 2
Π : tumeur pinéale
Σ : tumeur du s.e.d 3
B : tumeur basocellulaire 3
B0 : tumeur basocellulaire bénigne 1
B0A0 : naevus basocellulaire
B4 : tumeur basocellulaire bénigne avec dysplasie légère 2
B4A0 : naevomatose basocellulaire au cours du syndrome de gorlin
B4E0 : tumeur fibroepitheliale de pinkus
B7 : carcinome basocellulaire 14
β : tumeur blastemateuse 2
β0 : tumeur blastemateuse bénigne 5
B0F0 : tumeur blastemateuse benigne (sai)
B0F2 : blasteme persistant
B0F4 : blasteme massif
B0F9 : blastematose
B0G0 : diktyome
β7 : tumeur blastemateuse maligne 23


Glandularor/TubularDifferentiation
TherapeuticProcedure_PreventiveProcedure
NeoadjuvantTherapy
Cellular Differentiation
QuantitativeConcepts
Score
Interval
LessThan
GreatherThan
PercentUnit
SpatialConcepts
Area
QalitativeConcepts
Tubular
Residual
Present
Greater
Invasive
Indeterminate
ConceptualEntities
Formation
Activity
Comparison
Presentation
FunctionalConcept
LessThan
CellFunction
CellDifferentiationProcess
BodyPart_Organ_OrganComponent
Gland
NeoplasticProcess
Carcinoma
Invasive Carcinoma
Neoplasm
Finding
TumorMass
InvasiveLesion


NeoadjuvantTherapy
Cellular Differentiation
TherapeuticProcedure_PreventiveProcedure
Score
Interval
LessThan
QuantitativeConcepts
GreatherThan
PercentUnit
Area
SpatialConcepts
Tubular
Residual
Present
QalitativeConcepts
Greater
Invasive
Indeterminate
Formation
ConceptualEntities
Glandularor/TubularDifferentiation
Comparison
Activity
Presentation
FunctionalConcept
LessThan
Glandularor/TubularDiffe
All invasive breast carcinomas should be graded. The Nottingham combined histologic grade (Elston-Ellis modification of Scarff-Bloom-Richardson grading system) should be used for reporting. Within each stage grouping there is a relation between histologic grade and outcome.The Nottingham combined histologic grade evaluates the amount of tubule formation, the extent of nuclear pleomorphism, and the mitotic count (or mitotic rate). Each variable is given a score of 1, 2, or 3, and the scores are added to produce a grade.
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Glandularor/TubularDifferentiation
Link to: https://bioportal.bioontology.org/onto...
NeoadjuvantTherapy
Cellular Differentiation
TherapeuticProcedure_PreventiveProcedure
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QalitativeConcepts
Formation
ConceptualEntities
Activity
Comparison
Presentation
FunctionalConcept
LessThan
CellFunction
CellDifferentiationProcess
BodyPart_Organ_OrganComponent
Gland
NeoplasticProcess
Carcinoma
Invasive Carcinoma
Neoplasm
Finding
TumorMass
InvasiveLesion



Automatic Process of the Visual Representation (brings Sustainability)

AP Quantifiable Observations

- **Annotation Corpus**
  - 55/67 CAP protocols related to Malignant tumor analysis
  - 83 quantifiable Observations

- **Experts identification Content**

Result of identification of relevant terms and groups of terms by medical experts		
	Expert 1 (114)	Expert 2 (103)
Total Number of pertinent Terms	11	11
Total Number of « Group of Terms »	103	92

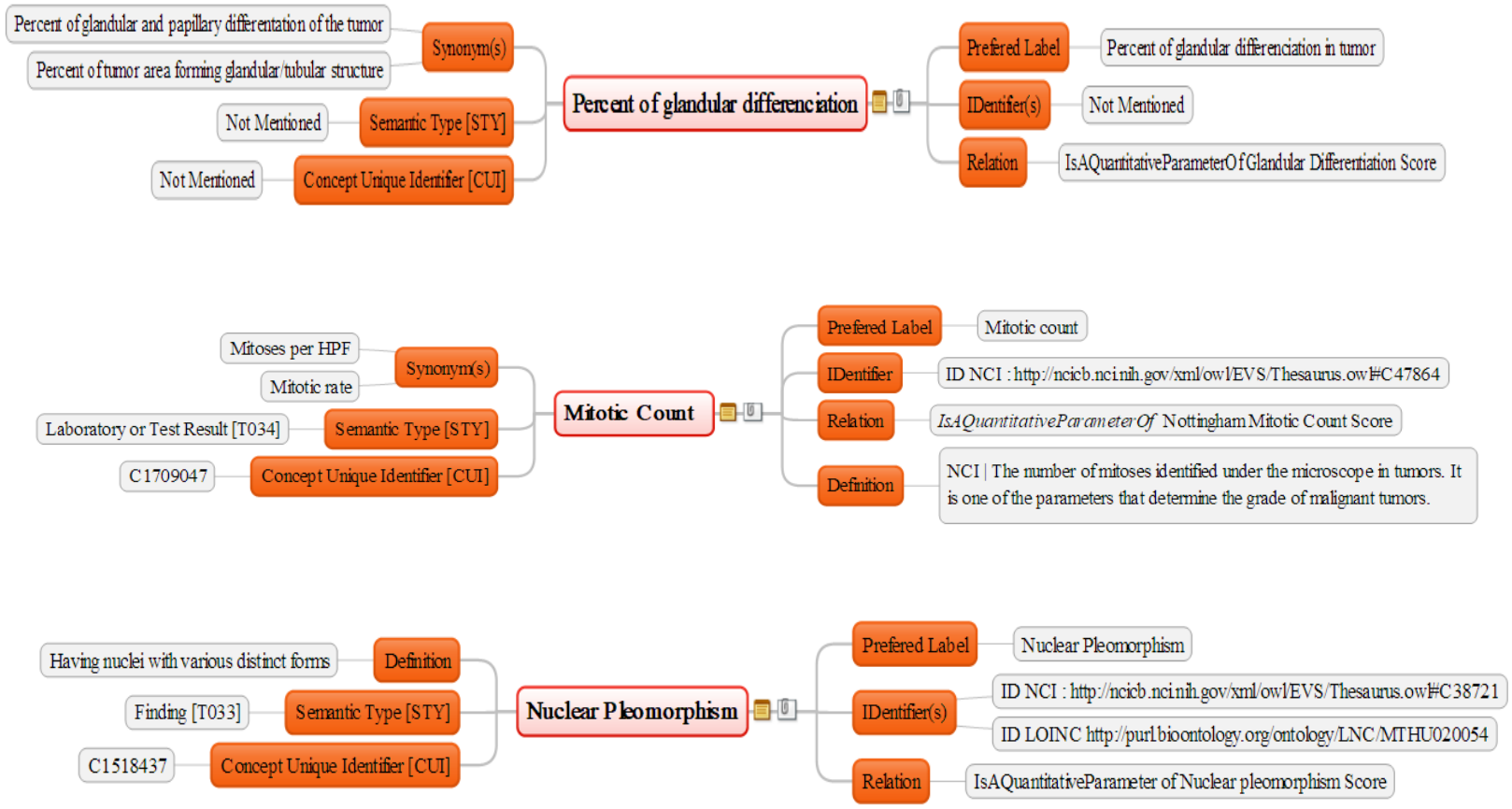
Inter-expert agreement analysis

- **F-measure**
  - 134 quantifiable parameters
  - 82 common to both experts
  - F-measure 76%
- **« Gold standard »**
  - 91 Quantifiable Parameters

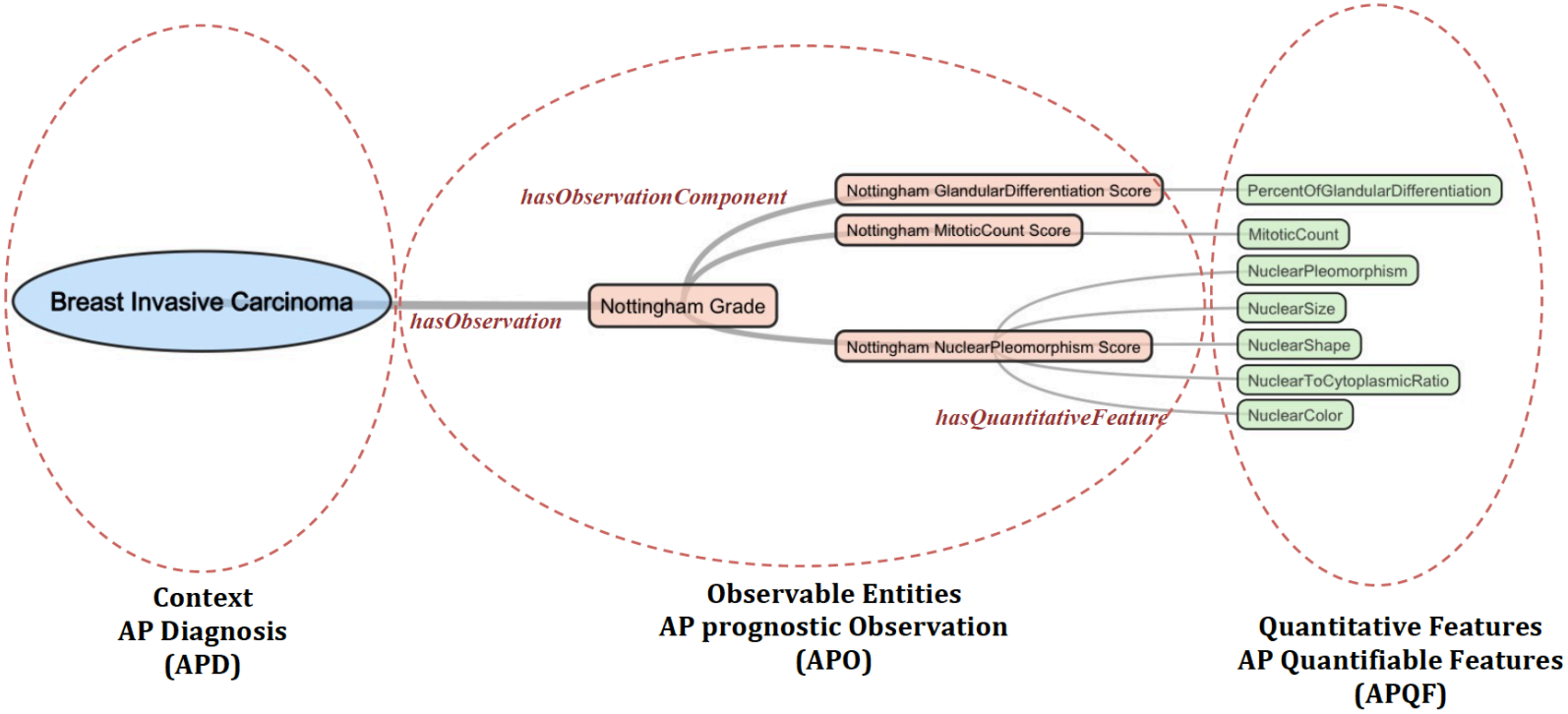
CAP Cancer Protocols	Quantifiable observation value set (NOTE)	Expert 2 (observation quantifiable par l'humain)	Expert 2 (parametre quantifiable par la machine*)
Esophagus_16Protocol_3200_final	Histologic Grade (Note D) Not applicable Grade X Cannot be assessed Grade 1 Well differentiated Grade 2 Moderately differentiated Grade 3 Poorly differentiated Grade 4 Undifferentiated D. Histologic Grade The histologic grades for esophageal squamous cell carcinomas are as follows: Grade X Grade cannot be assessed Grade 1 Well differentiated Grade 2 Moderately differentiated Grade 3 Poorly differentiated If there are variations in the differentiation within the tumor, the highest (least favorable) grade is recorded. In general, mucoepidermoid carcinoma and adenoid cystic carcinoma of the esophagus are not amenable to grading. For adenocarcinomas, a suggested grading system based on the proportion of the tumor that is composed of glands is as follows: Grade X Grade cannot be assessed Grade 1 Well differentiated (greater than 95% of tumor composed of glands) Grade 2 Moderately differentiated (50% to 95% of tumor composed of glands) Grade 3 Poorly differentiated (49% or less of tumor composed of glands) Undifferentiated tumors cannot be categorized as squamous cell carcinoma or adenocarcinoma (or other) type. They are classified as "undifferentiated carcinomas" in the WHO classification of tumor types (see above) and may be assigned grade 4. Small cell carcinomas are not typically graded but are high-grade tumors and would correspond to grade 4. The revised TNM staging system for esophageal carcinomas incorporates tumor grade and histologic type in the stage groupings (see Note H). For purposes of staging, grade 4 carcinomas (undifferentiated carcinomas) are staged as grade 3 squamous cell carcinomas.1 Grade X tumors are grouped as grade 1 carcinomas.	Differentiation carcinoma Grade X Grade cannot be assessed Grade 1 Well differentiated Grade 2 Moderately differentiated Grade 3 Poorly differentiated Grade 4 Undifferentiated carcinomas Proportion of glands formation*	proportion of gland formation
Esophagus_16Protocol_3200_final	Treatment Effect (applicable to carcinomas treated with neoadjuvant therapy) (select all that apply) (Note G) No prior treatment Present No viable cancer cells (complete response, score 0) Single cells or rare small groups of cancer cells (near complete response, score 1) Residual cancer with evident tumor regression, but more than single cells or rare small groups of cancer cells (partial response, score 2) Extensive residual cancer with no evident tumor regression (poor or no response, score 3) Treatment history not known G. Treatment Effect Response of tumor to previous chemotherapy or radiation therapy should be reported. Several systems for tumor response have been advocated, and a modified Ryan scheme is suggested, which has been shown to provide good interobserver reproducibility provide prognostic significance in rectal cancer.11 Modified Ryan Scheme for Tumor Regression Score11 Description Tumor Regression Score	score 0 complete response score 1 near complete response score 2 partial response score 3 poor or no response Treatment history not known Count of viable cancer cells* Ryan scheme Residual tumor	Number of viable cancer cells

Results of the BioPortal ontologies giving the best coverage rate for AP quantifiable parameters

Gold Standard		Reference Ontologies (Coverage %)				
Type of Cancer	# of Concepts	NCIT	SNOMED CT	LOINC	RADLEX	PATHLEX
Colon & rectum	6	94%	48%	39%	38%	31%
Œsophagus	20	75%	41%	51%	28%	17%
Prostate	13	85%	61%	49%	7%	27%
Breast	66	70%	52%	54%	26%	15%
Melanoma	19	66%	51%	51%	22%	9%
Average Coverage/Ontologie		78%	50%	49%	24%	20%

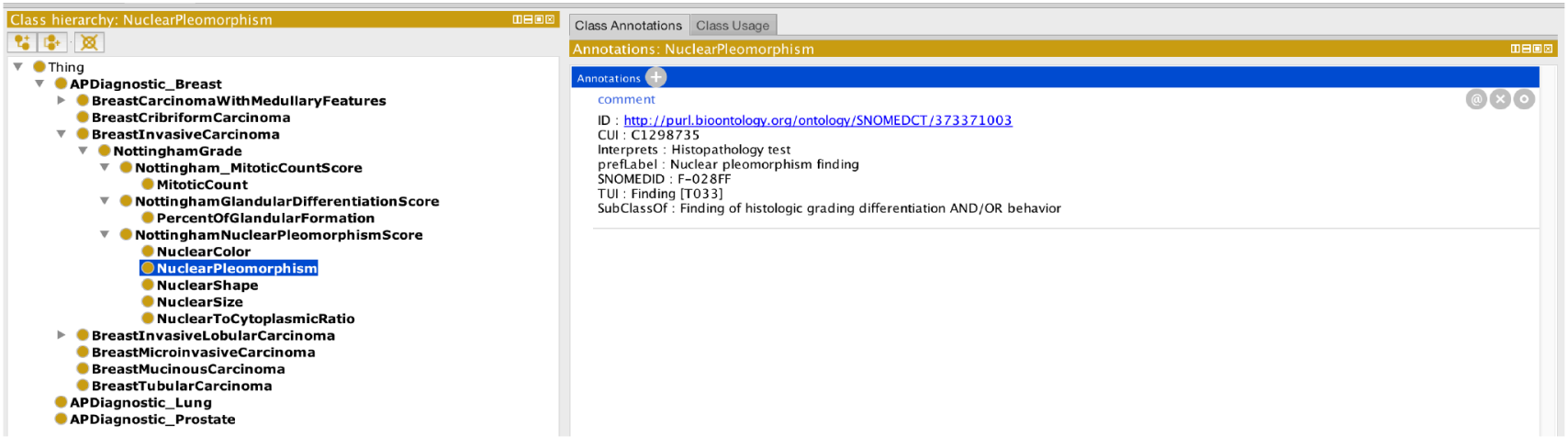


Mental Maps in the context of Breast Cancer



APD, APO and APQF organisation in the context of Breast Invasive Carcinoma prognosis

AP Diagnostic (APD)		AP prognostic Observations (Grades, Prognostics scores, Treatment Effect evaluation) (APO)					AP Quantitative Features (APQF)	
Class	**Relation**	**Class**	**Description**	**Relation**	**Class**	**Description**	**Relation**	**Class**
Breast Invasive Carcinoma	*hasObservation*	**Nottingham Grade**	The Nottingham combined histologic grade evaluates the amount of tubule formation, the extent of nuclear pleomorphism, and the mitotic count. Grade 1 (scores of 3, 4, or 5) Grade 2 (scores of 6 or 7) Grade 3 (scores of 8 or 9)	*hasObservationComponent*	**Nottingham Glandular Differentiation Score**	Score 1 (>75% of tumor area forming glandular/tubular structures) Score 2 (10% to 75% of tumor area forming glandular/tubular structures) Score 3 (<10% of tumor area forming glandular/tubular structures)	*hasQuantitativeFeature*	Percent of glandular differentiation
				hasObservationComponent	**Nottingham MitoticCount Score**	Score 1 (≤3 mitoses per mm) Score 2 (4-7 mitoses per mm) Score 3 (≥8 mitoses per mm)	*hasQuantitativeFeature*	Mitotic count
				hasObservationComponent	**Nottingham NuclearPleomorphism Score**	Score 1 (nuclei small with little increase in size in comparison with normal breast epithelial cells, regular outlines, uniform nuclear chromatin, little variation in size) Score 2 (cells larger than normal with open vesicular nuclei, visible nucleoli, and moderate variability in both size and shape) Score 3 (vesicular nuclei, often with prominent nucleoli, exhibiting marked variation in size and shape, occasionally with very large and bizarre forms)	*hasQuantitativeFeature*	NuclearPleomorphism
								NuclearSize
								NuclearShape
								Nuclear-to-cytoplasmic ratio
								NuclearColor



Proposal of an organ dependent hierarchical organization of APQF taking into account the Breast AP diagnostic context



Proposal of an organ independent hierarchical organization of APQF taking into account generic quantifiable features

Top 5 BioPortal **unrestricted** "biomedical ontologies"

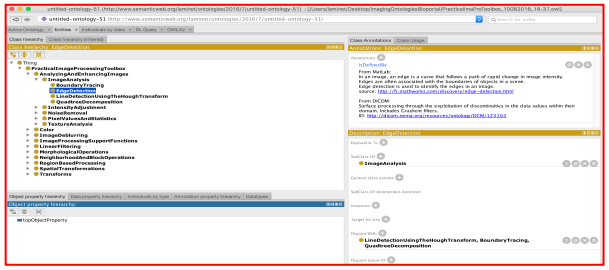
#	Name	Category	Classes
1	Logical Observation Identifier Names and Codes (LOINC)	Health	187123
2	Material Rock Igneous (MATRROCKIGNEOUS)	Upper Level Ontology	3535
3	Medical Subject Headings (MESH)	Health	261990
4	Material Natural Resource (MNR)	Upper Level Ontology	3554
5	National Cancer Institute Thesaurus (NCIT)	Vocabularies	118941

The coverage results with Corpus#1 were :

57.7% for single ranked ontology (NCIT)

75.2% for ontology sets (NCIT, SNOMEDCT, SWEET and LOINC)

Formal world



Practical Image Processing Tasks Ontology - PIPTO

Visualisation

OWL to OWL/RDF

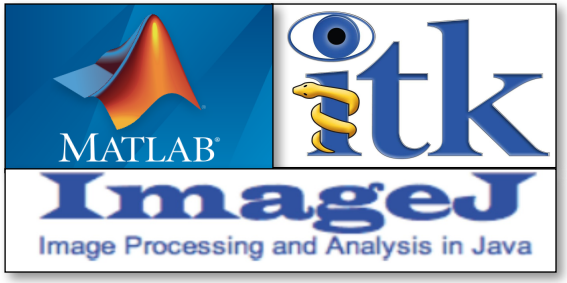
Identified Concept Number

- Matlab: **565**
- ITK: **348**
- ImageJ: **259**

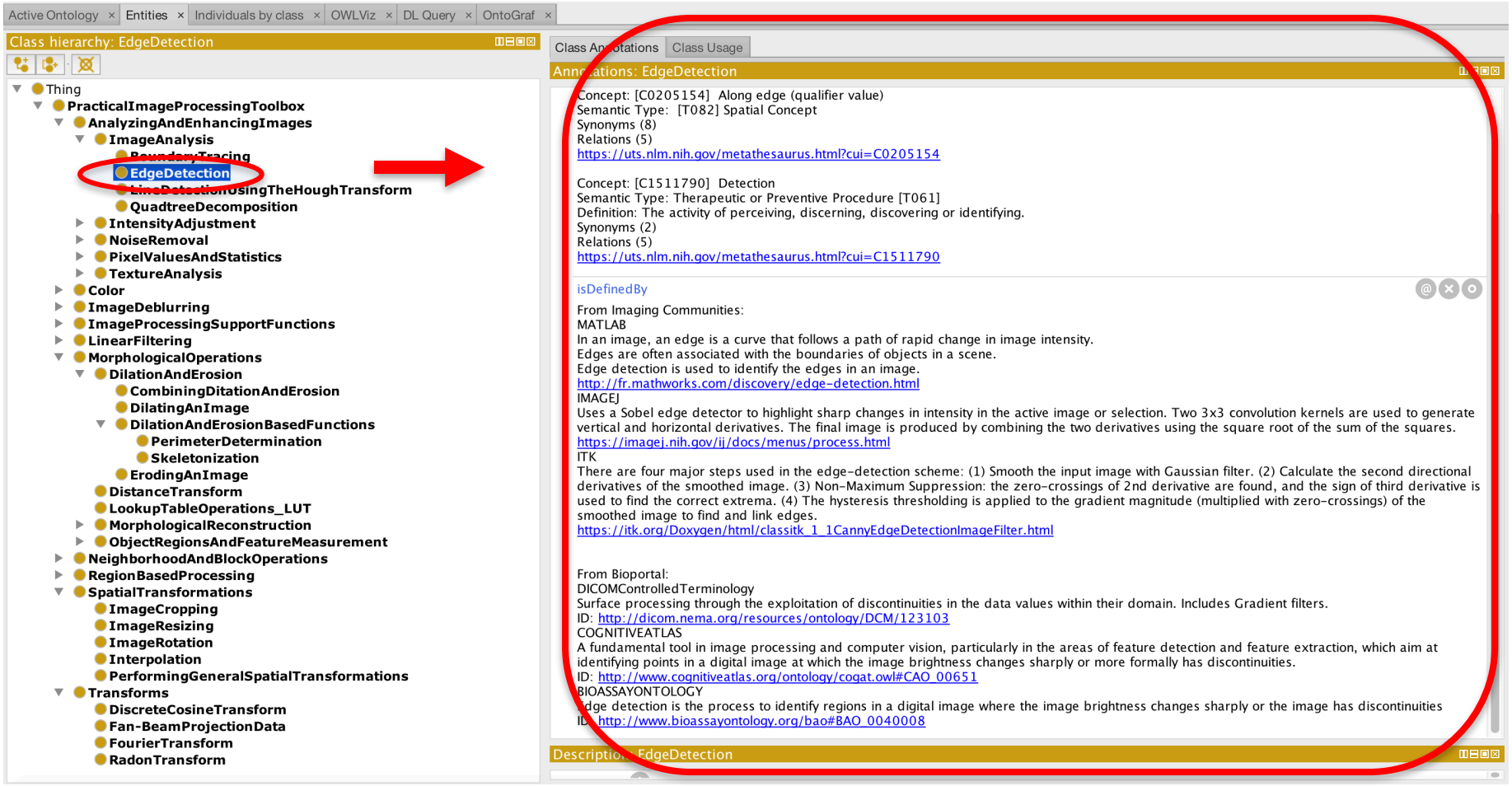


TXT to OWL

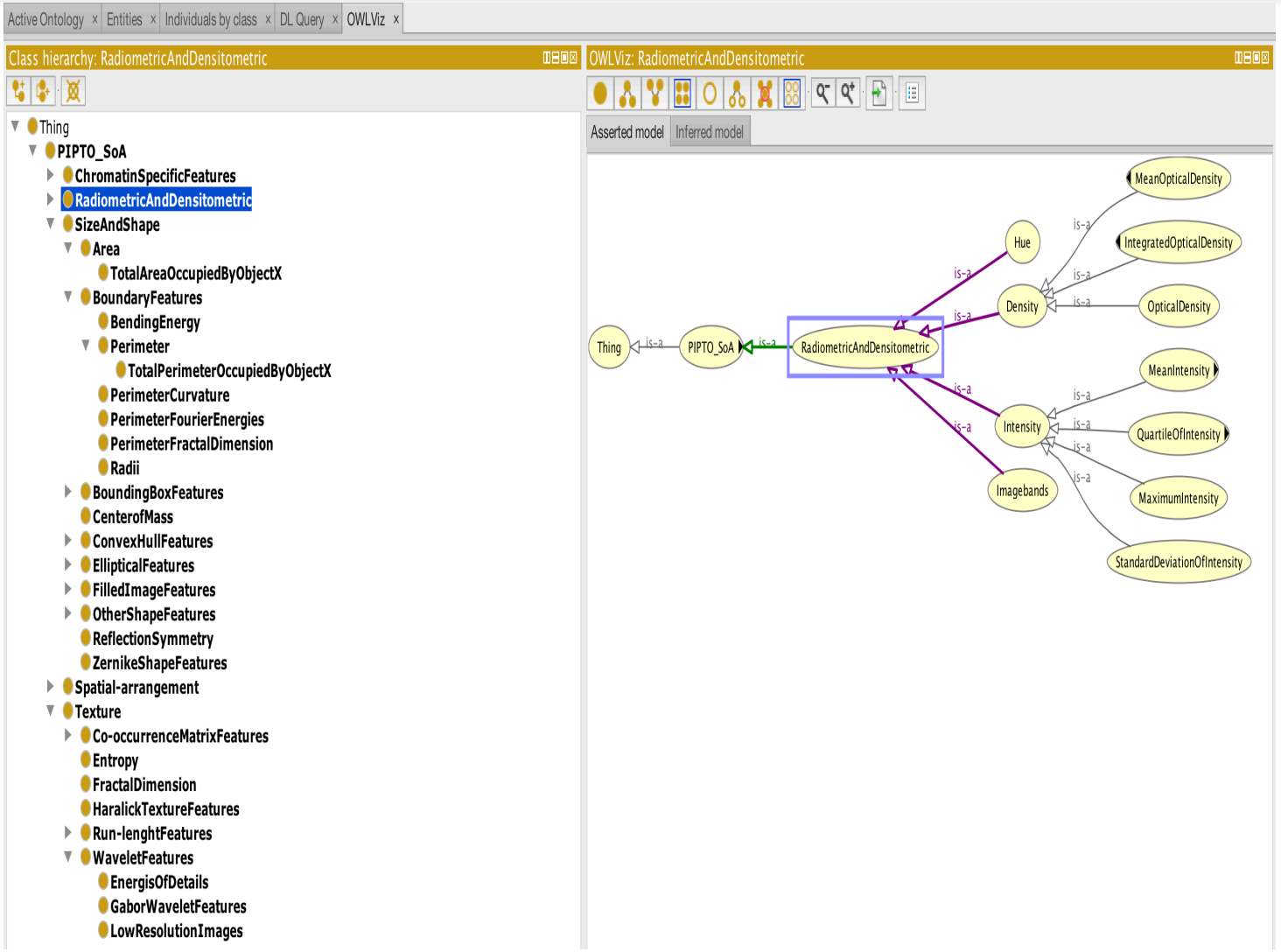
Operational world



Sources from (3) Image Analysis Communities



PIPTO issued from imaging community software libraries



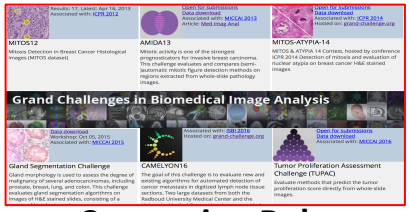
Practical Image Processing Task Ontology (PIPTO) issued from the State Of the Art

Bridging the semantic gap between histopathology and imaging domains

Formal world

Visualisation

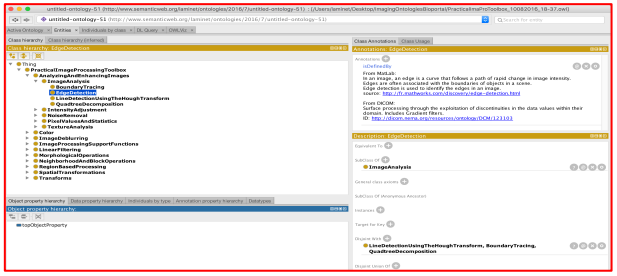
Operational world



Analysis Rules

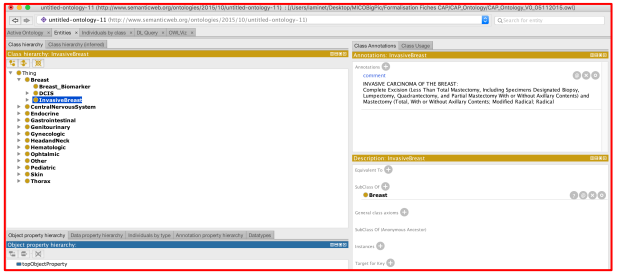
Conversion Rules

Analysis Rules



APQF

Conversion Rules



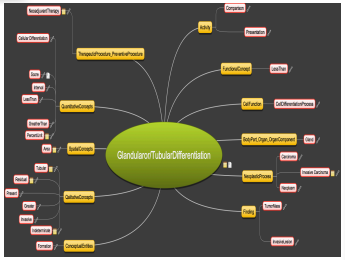
PIPTO

SVG to OWL/R DF

OWL to OWL/R DF

Breast cancer use-case:

- **23** concepts from the
- **5** different ontologies
- **11** UMLS STY





Imaging concepts

- Matlab: **565**
- ITK: **348**
- ImageJ: **259**

XML to SVG

TXT to OWL

- **66** CAP protocols
- **20** organ specific
- **133** UMLS STY
- **54** relations
- **580** ontologies in Bioportal



COLLEGE of AMERICAN PATHOLOGISTS

MATLAB

itk

ImageJ

Image Processing and Analysis in Java

DICOM

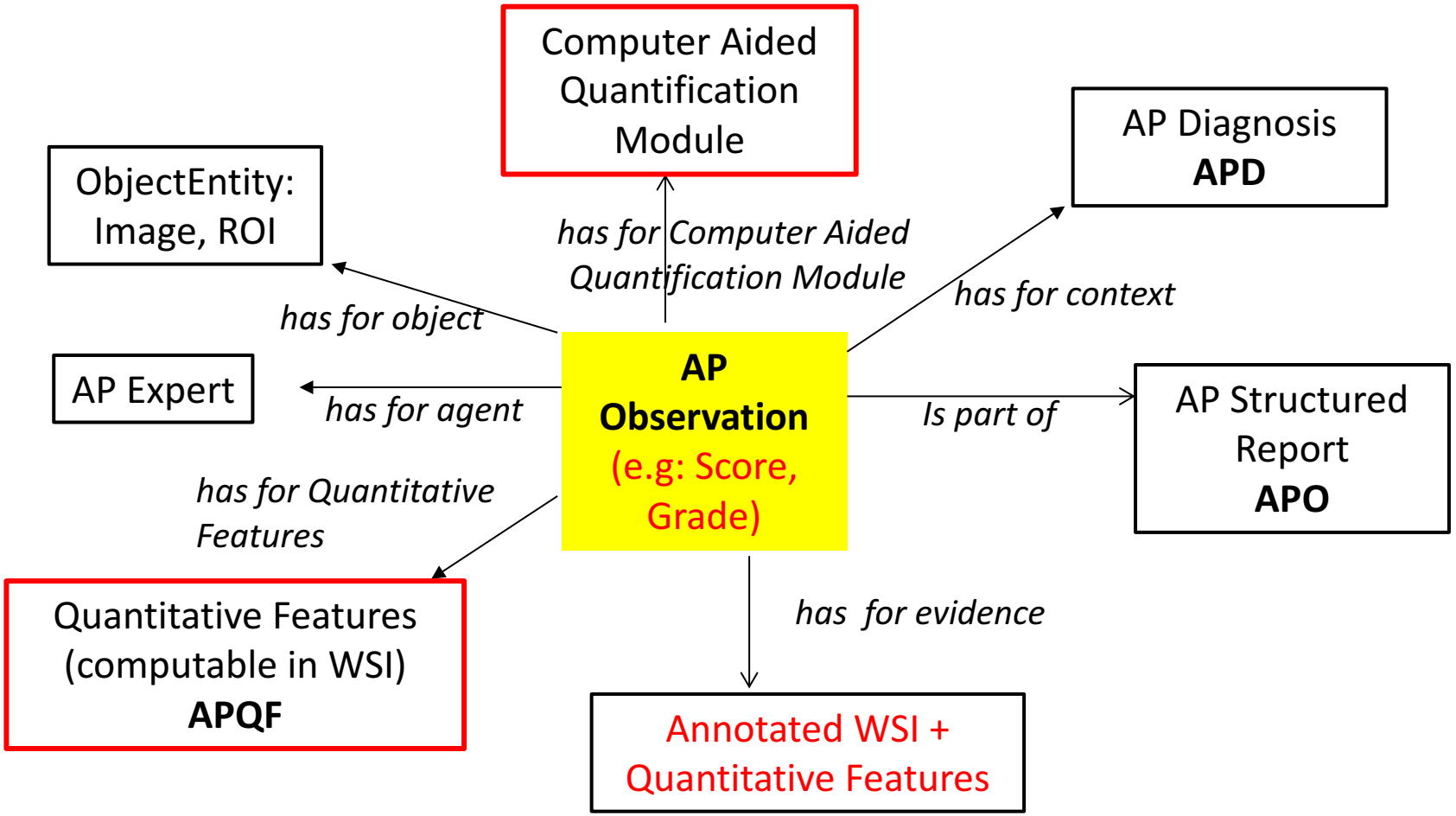
Digital Imaging and Communications in Medicine

Anatomopathology Domain

Image Processing Domain

APQF	PIPTO (module)
Feature Category	**CellProfiler Module**
Pixel Correlation of the cells	MeasureCorrelation
Pixel Correlation of the nuclei	MeasureCorrelation
Image granuarity	MeasureGranularity
Area occupied by cells	MeasureImageAreaOccupied
Area occupied by nuclei	MeasureImageAreaOccupied
Image intensity	MeasureImageIntensity
Image quality	MeasureImageQuality
Pixel intensity of the cell	MeasureObjectIntensity
Pixel intensity of the nuclei	MeasureObjectIntensity
Neighboring relation of the cells	MeasureObjectNeighbors
Neighboring relation of the nuclei	MeasureObjectNeighbors
Radial distribution of pixel intensity of the cell	MeasureObjectRadialDistribution
Radial Distribution of the Pixel Intensity of the Nuclei	MeasureObjectRadialDistribution
Cell size and shape	MeasureObjectSizeShape
Nucleus size and shape	MeasureObjectSizeShape
Cell texture	MeasureTexture
Nucleus texture	MeasureTexture

Matching low-level AP Quantifiable features to Specific imaging Tasks For Score prognostic evaluation



Bridging the Gap for a score/grade prognostic evaluation





Computer Aided Quantification Module (PIPTO):

- MeasureCorrelation (e.i: Correlation coefficients between hematoxylin and eosin stained nucleus regions)
- MeasureImageAreaOccupied
- MeasureObjectIntensity (e.i: cell, nuclei)
- MeasureObjectSizeShape

AP Expert: Dr CD

has for agent

has for Computer Aided Quantification Module

BreastInvasiveCarcinoma (APD)

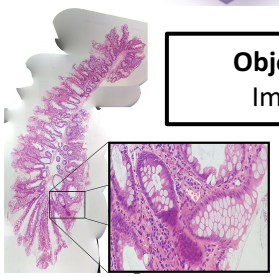
has for context

Notthingham histologic score

Nuclear Pleomorphism

Is part of

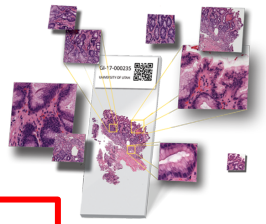
AP Structured Report (APO)+Annotated WSI:
Score1, 2 or 3
(e.i: Score2: cells larger than normal with, open vesicular nuclei, visible nucleoli, and moderate variability in both size and shape)



ObjectEntity: Image, ROI

has for (raw) data/images

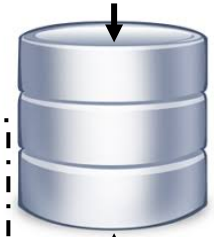
has for QuantitativeFeatures



Quantitative Features (APQF):

- Pixel Correlation of the nuclei
- Area occupied by nuclei
- Pixel intensity of the nuclei
- Cell size and shape
- Nucleus size and shape

AP Data Warehouse:
Semantically enriched AP Report
Annotated WSI+AP Quantitative Features (evidences)



Example of Nottingham Nuclear Pleomorphism Score prognostic evaluation

5. Concluding Remarks

Contributions (1/3)

- The development of **two standard-based terminological systems** in the **AP domain** to

  →bridge the semantic gap between **diagnostic histopathology** and **image analysis**

- The **scientific state-of-the-art** in the fields of Medical Informatics, Image analysis, Information Systems, and Biomedical Engineering.

Contributions (2/3)

- We proposed a **semi-automated workflow** for **selecting candidate ontologies/semantic sources** for semantic annotation of textual documents in a given domain.

  →this workflow was **applied on the AP** Quantifiable Features (APQF).

- Proposed an **Approach, Tool** (Mental Maps) and **Formal representation** based on the CAP-CC&Ps,

  →to support AP experts in building a standard-based **representation of low-level morphological abnormalities.**

Contributions (3/3)

- We **built a formal model of AP** Quantifiable Features (**APQF**) in which concepts are organized

  →by **feature categories** and defined in the **context** of each **organ specific** grade/score system.

- We **identified key imaging knowledge** and **concepts** issued from different community sources**: Matlab, ImageJ, ITK** and **histopathology imaging contests**.

  → initiate a formal model **PIPTO** by integrating this knowledge with **existing semantic resources** in **NCBO** and **UMLS**.

Publications

Traoré L, Kergosien Y, Racoceanu D, “Bridging the Semantic Gap Between Diagnostic Histopathology and Image Analysis,” **Medical Informatics Europe** Stud. Health Technol. Inform., vol. 235, pp. 436–440, **2017**.

Traoré L, Daniel C, Jaulent MC, Schrader T, Racoceanu D, Kergosien Y “Modélisation sémantique d'un outil d'exploration et d'analyse d'images histopathologiques”, Oral communication to 1st Forum Franco-Québécois d’innovation en Santé 11-**12 Oct. 2016**, **Montréal**

Traoré L, Daniel C, Jaulent MC, Schrader T, Racoceanu D, Kergosien Y "A sustainable visual representation of available histopathological digital knowledge for breast cancer grading" **Diagnostic Pathology Journal**, vol. 2, no.1, **Jun. 2016**

Declaration of Invention, UPMC's Directorate for Research & Technology Transfer (**DGRTT**) office - ongoing

Limits

- Proposed taxonomic organizations & hierarchy are subject to the **validation** of **domain specific experts and organizations** (by Organ for AP, by Task for imaging)

- **Greater participation** of the AP community is needed in the **development**, **adoption**, and **maintenance** of such a source in a **sustainable** manner

Perspectives (1/2)

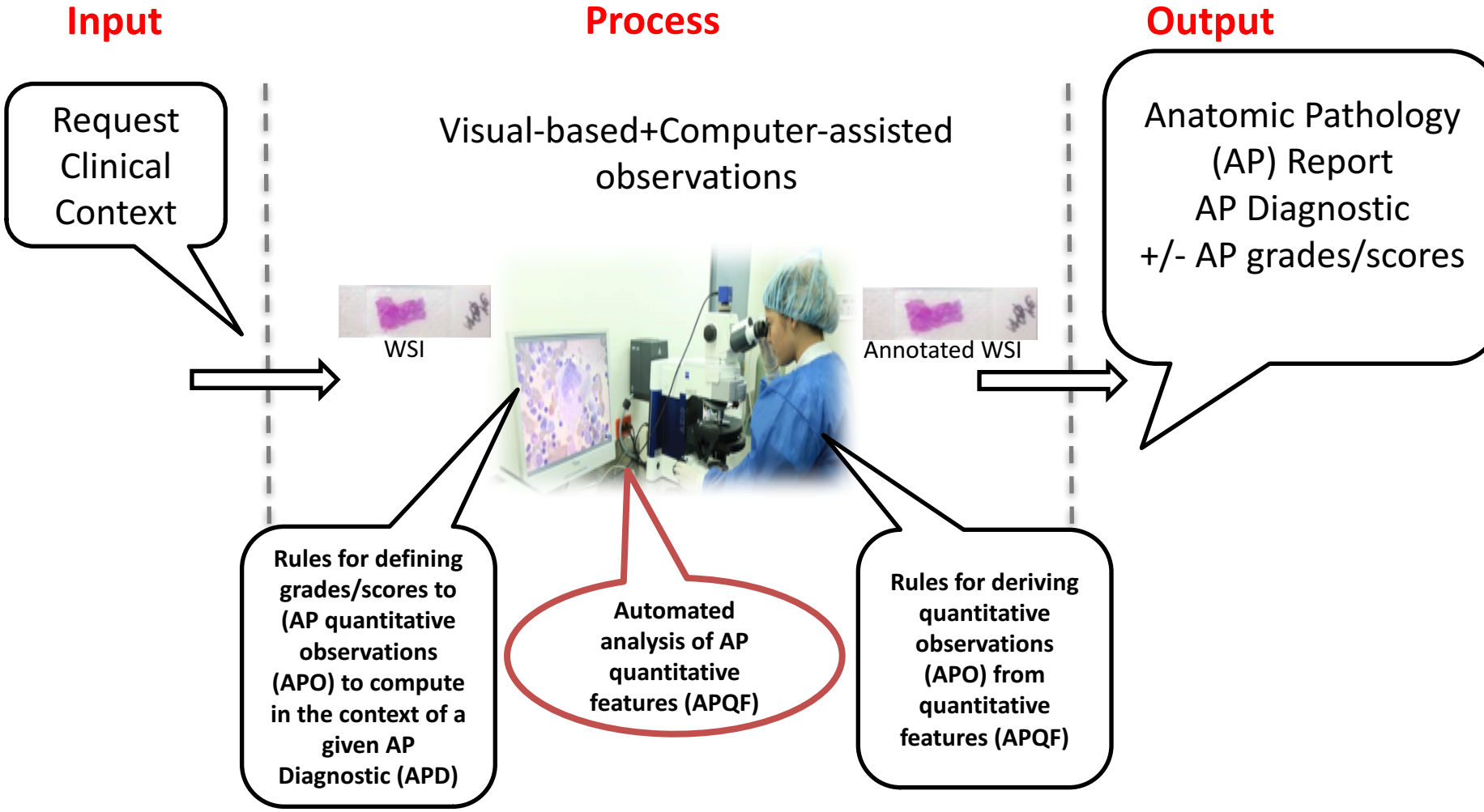
Considering **high-level ontologies** such as
- **BFO** (Basic Formal Ontology)
- **DOLCE** (Descriptive Ontology for Language and Cognitive Engineering)

→ Provide the basic entities and relationships for a **better overall coherence**.

Consolidation of the semantic modeling (properties, relations, rules) to
→ Carry out **reasonings** within the framework of the project Smart'GRADE

Integrate **CAP biomarker protocols** as these certainly
→ Play a **crucial role** in diagnostic or prognostic **decision-making**

Perspectives (2/2)



Scenario of a computer-assisted process in the era of digital pathology for the exploration of a WSI in an Anatomic Pathology Laboratory







Ecole doctorale Pierre Louis de santé publique
ED 393 Epidémiologie et Sciences de l'Information Biomédicale












Thanks

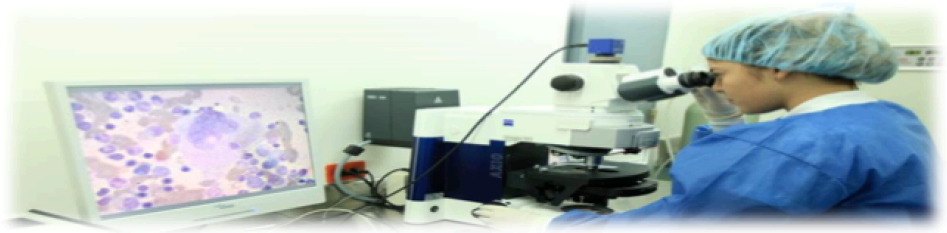
Identifier & **Extraire** les observations ACP dans les fichiers CAP

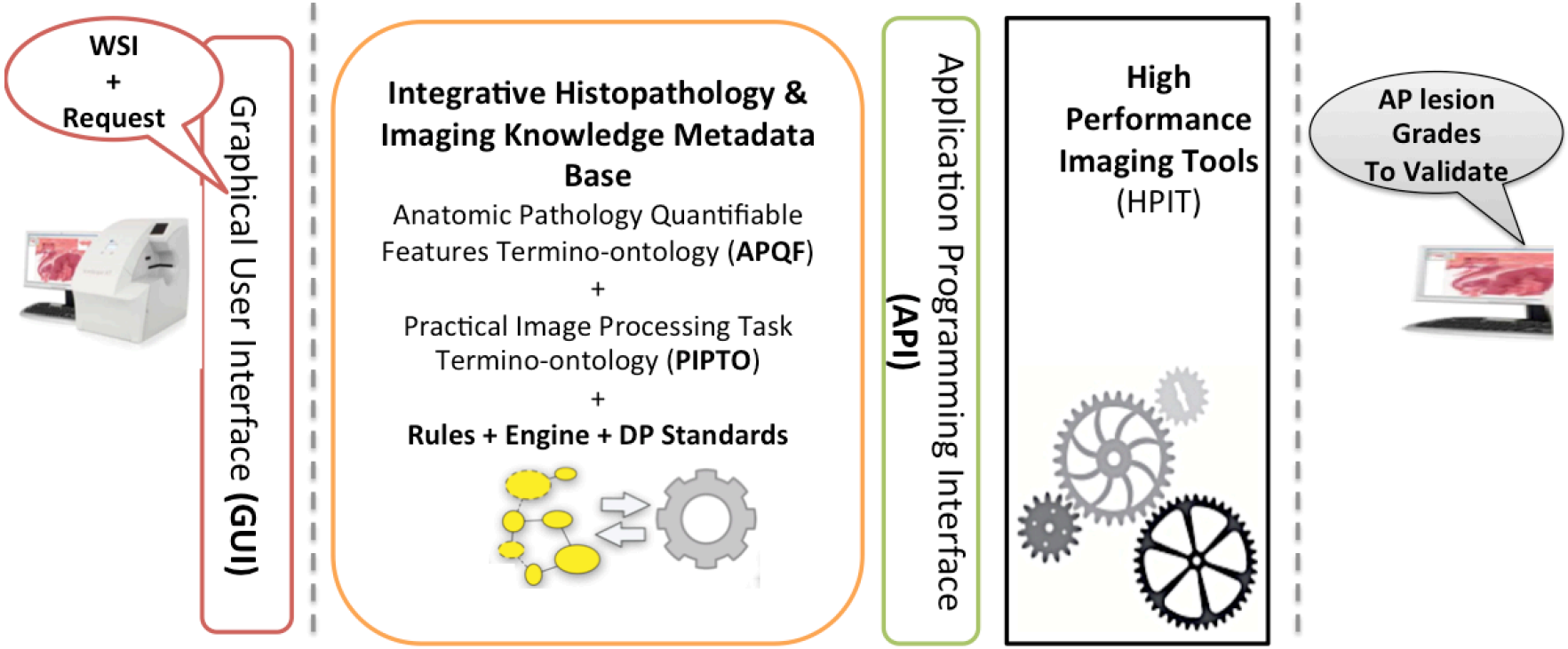
67 Protocoles : **55** protocoles: **83 Questions Quantifiables** + « **Gold Standard**»

12 protocoles: NO Quantifiable Questions

1. BoneMarrow_12protocol_3011
2. HodgkinLymphoma_13protocol_3100
3. Mesothelioma_12protocol_3100_PasDeGrade
4. NonHodgkinLymph_13protocol_3200
5. OcularAdnexa_12protocol_3000
6. PlasmaCell_15Protocol_1000
7. Testis_13protocol_3300_PasDeGrade
8. Thymus_12protocol_3100
9. Thyroid_16Protocol_3200_final
10. Trophoblast_15Protocol_3100_final_PasDeGrade
11. UvealMelanom_16Protocol_3300_Final_PasDeGrade
12. Wilms_12protocol_3102_PasDeGrade

Order Process Report





Main Components of Smart'Grade Computer Aided quantification platform for AP laboratory

Finding reference ontologies

Results of the BioPortal ontologies giving the best coverage rate for AP quantifiable parameters						
Gold Standard		Reference Ontologies (Coverage %)				
Type of Cancer	# of Concepts	NCIT	SNOMED CT	LOINC	RADLEX	PATHLEX
Colon & rectum	6	94%	48%	39%	38%	31%
Œsophagus	20	75%	41%	51%	28%	17%
Prostate	13	85%	61%	49%	7%	27%
Breast	66	70%	52%	54%	26%	15%
Melanoma	19	66%	51%	51%	22%	9%
Average Coverage/Ontologie		78%	50%	49%	24%	20%

Top 5 Bioportal **restricted** to **"imaging category"** ontologies

#	Name	Category	Classes
1	Radiation Oncology Ontology (ROO)	Development, Health, Human, Imaging, Vocabularies	1183
2	DICOM Controlled Terminology (DCM)	Imaging	3476
3	Information Artifact Ontology (IAO)	Biomedical Resources, Imaging, Other	180
4	Biomedical Informatics Research Network Project Lexicon (BIRNLEX)	Anatomy, Imaging	3580
5	Neural ElectroMagnetic Ontology (NEMO)	Anatomy, Biological Process, Experimental Conditions, Human, Imaging	1851

The coverage results with Corpus#1 were :

9.0% for single ranked ontology

21.8% for ontology sets